THE PULLBACK OF A QUASI-COHERENT MODULE IS QUASI-COHERENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The reduction of the statement to a module admitting a global presentation by direct sums of copies of the structure sheaf. Obtained by the same argument as in the proof of the author lemma lemma-pullback-locally-generated, invoked at source lines 1311-1313. Reduces quasi-coherence to the case where the module is presented by two direct sums.
- The compatibility of the pullback with colimits and its right exactness. Invoked inside the counted proof at source lines 1314-1315 as a pointer to a lemma of this chapter. Transports the presentation to the pullback.

Source reference. Original lemma lemma-exactness-pushforward-pullback: sources/stacks-modules/pinned-original-source.tex, source lines 990–990. Navigation pointer only: the counted text refers to the Stacks item lemma-exactness-pushforward-pullback, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-pullback-locally-generated: sources/stacks-modules/pinned-original-source.tex, source lines 1045–1045. Navigation pointer only: the counted text refers to the Stacks item lemma-pullback-locally-generated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{G}$ of a quasi-coherent $\mathcal{O}_Y$-module is quasi-coherent.*

Proof. Arguing as in the proof of Lemma lemma-pullback-locally-generated we may assume $\mathcal{G}$ has a global presentation by direct sums of copies of $\mathcal{O}_Y$. We have seen that f^* commutes with all colimits, and is right exact, see Lemma lemma-exactness-pushforward-pullback. Thus if we have an exact sequence

$$\bigoplus\nolimits_{j\in J} \mathcal{O}_Y \longrightarrow \bigoplus\nolimits_{i\in I} \mathcal{O}_Y \longrightarrow \mathcal{G} \longrightarrow 0$$

then upon applying f^* we obtain the exact sequence

$$\bigoplus\nolimits_{j\in J} \mathcal{O}_X \longrightarrow \bigoplus\nolimits_{i\in I} \mathcal{O}_X \longrightarrow f^*\mathcal{G} \longrightarrow 0.$$

This implies the lemma. □